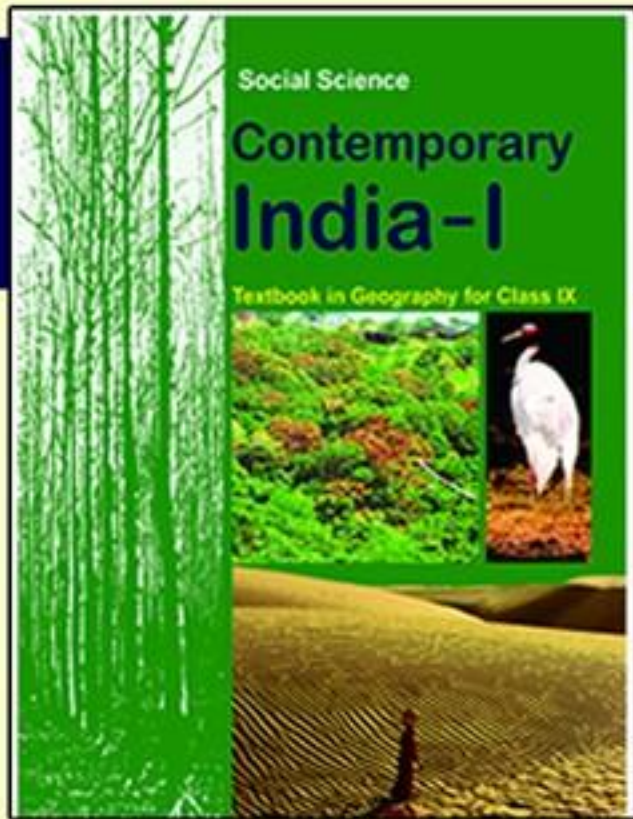




Learning **Space**



# NCERT BASICS

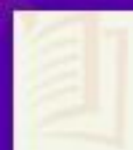


## GEOGRAPHY

THE UNIVERSE AND OUR  
SOLAR SYSTEM

**MODULE - 1**





## GEOGRAPHY ...

### THE EARTH



### DESCRIPTION



### ERATOSTHENES



The Geography is not that simple and it is **multi-dimensional** concept.





UNIVERSE ...  
BILLIONS OF GALAXIES



GALAXY...  
BILLIONS OF STARS



This is our Milky Way Galaxy!

STAR...  
WITH SEVERAL PLANETS



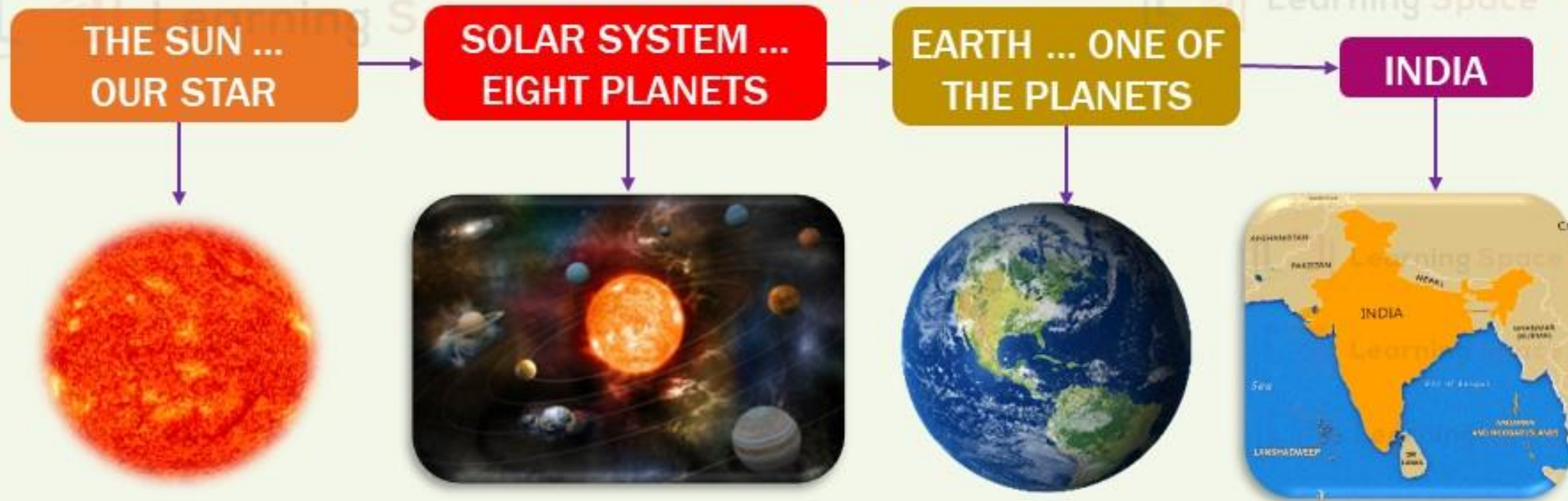
The Sun – Our Star!

### WHAT IS A GALAXY?

It is a huge collection of gas, dust, and billions of stars and their solar systems.

Canis Major Dwarf Galaxy -  
The known closest galaxy  
to us!



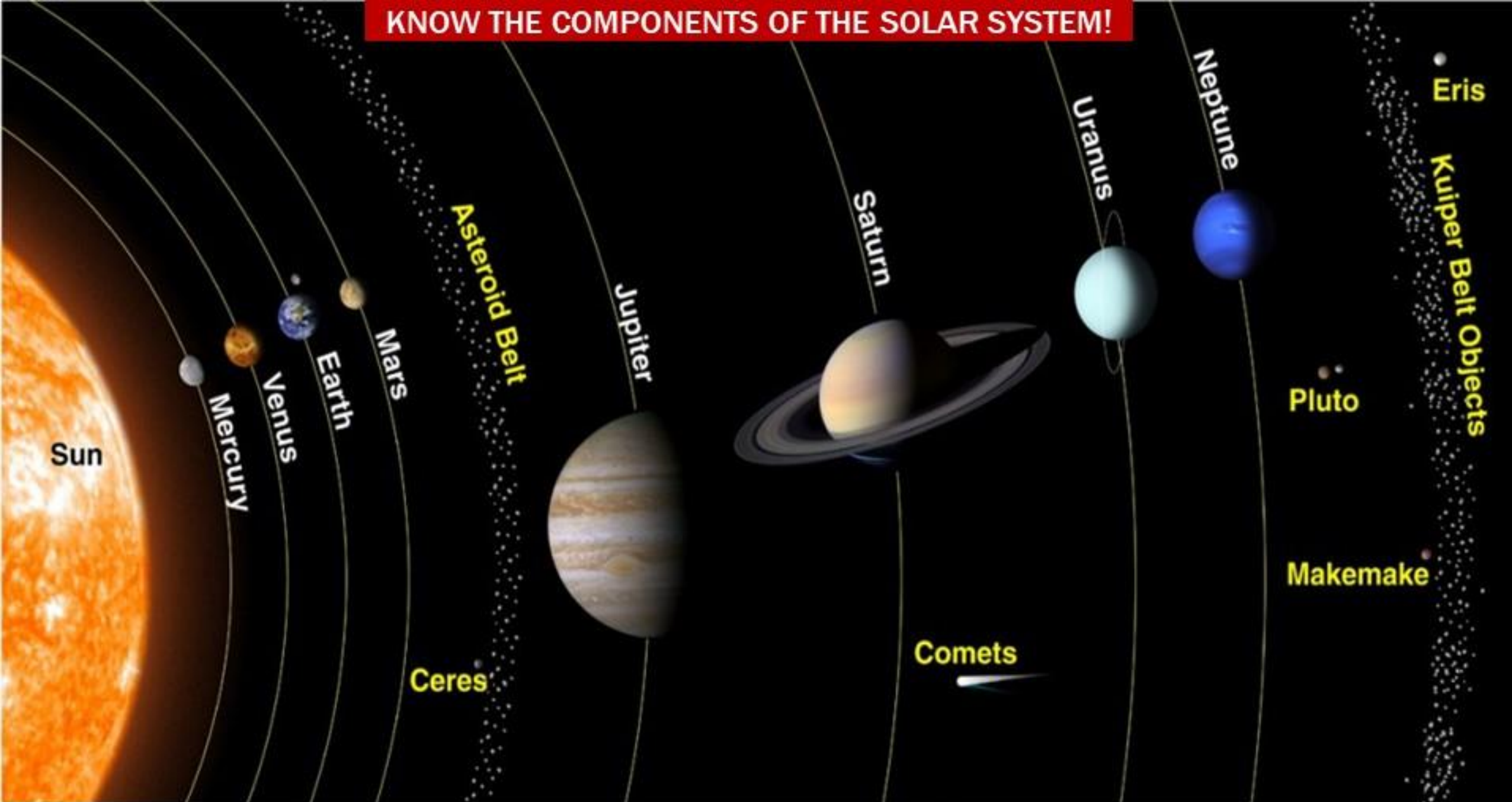


## KNOW MORE ABOUT THE SUN

- Made up of Hydrogen - 91.2%, Helium - 8.8%.
- Accounts for 99.85% of mass of the solar system.
- Continuously gives off energy in the form of electromagnetic radiation.



## KNOW THE COMPONENTS OF THE SOLAR SYSTEM!



**Satellite:** Any moon, planet or a machine in space that moves around a planet or a star.

## SO MANY OBJECTS... WHY ONLY 8 PLANETS?



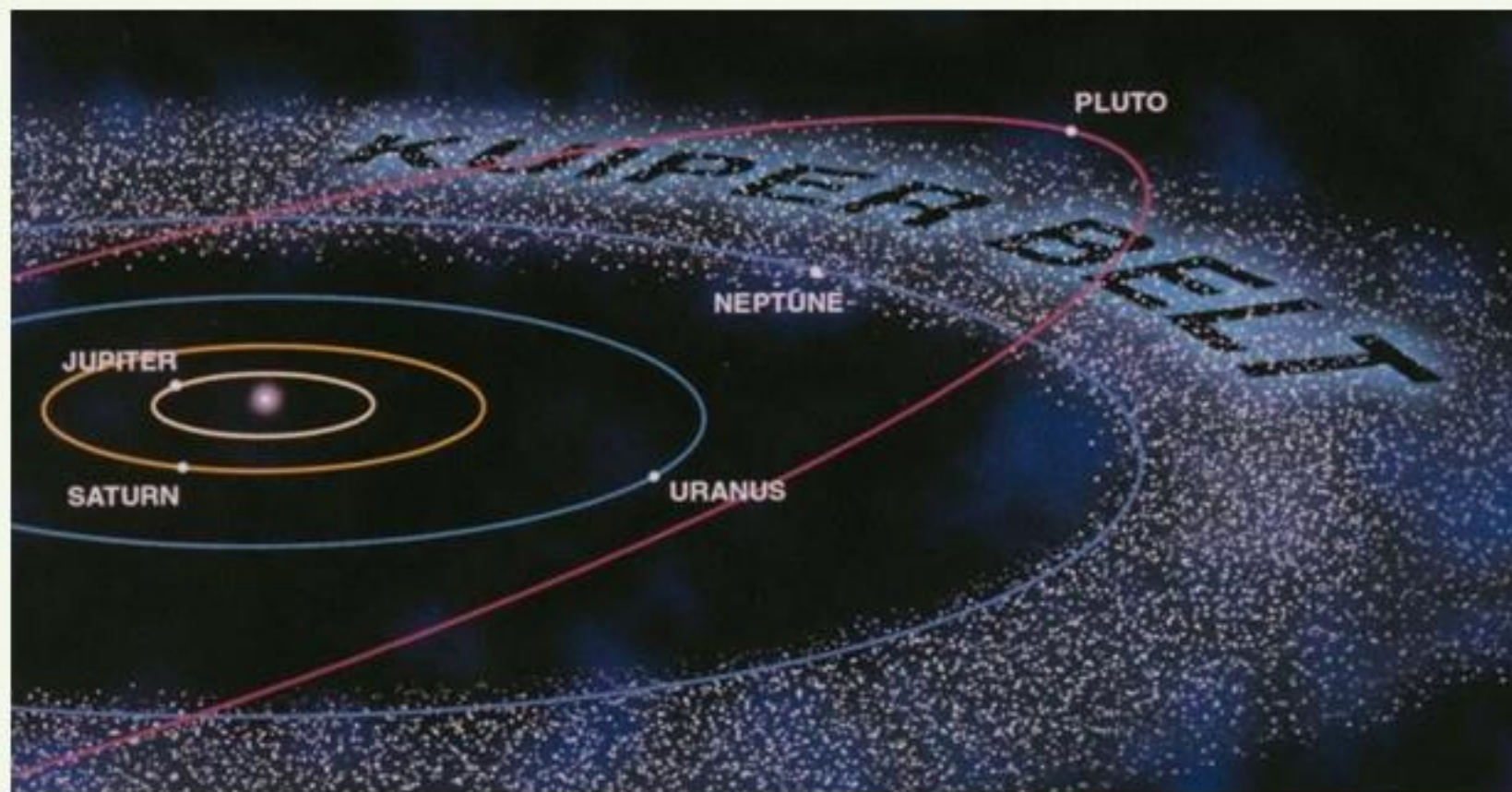
- Pluto was derecognised as a dwarf planet in 2006, as it could not satisfy the condition no. 3.
- It is in the Kuiper belt.

## TO BECOME A PLANET

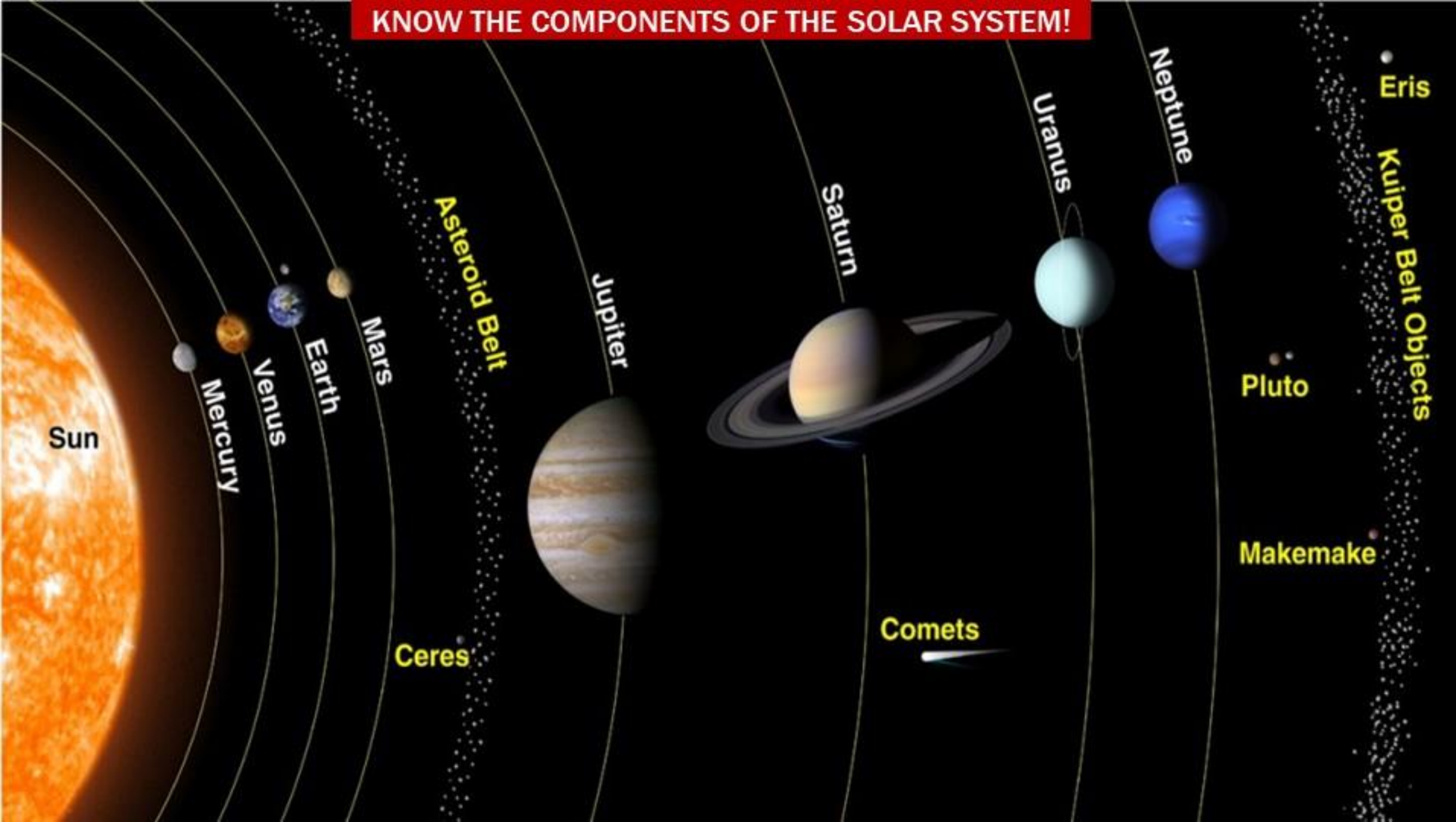
- The three conditions of the IAU for a full-sized planet are:
  1. It must be in orbit around the Sun.
  2. It must be large enough that it takes on a nearly round shape.
  3. It must have “cleared the neighborhood” around its orbit.

## WHAT DOES THE KUIPER BELT CONTAIN?

- The inner edge of the Kuiper Belt **begins** at the orbit of Neptune.
- Larger than 100 km across.
- It is the region of icy bodies, dwarf planets.



# KNOW THE COMPONENTS OF THE SOLAR SYSTEM!





## ASTERIOD BELT

Formed probably due to the reason that not all the dust and rocks were pulled together by gravity into planets.

## ASTERIODS

- They are the numerous rocky and airless **leftovers** which move around the Sun. Also called Minor Planets or Planetoids.
- These objects larger than 10 meters.

## CERES WAS THE LARGEST ASTERIOD... BUT ITS NOT SO NOW!

- In the year 2006 ceres was promoted as a dwarf planet by the International Astronomical Union (IAU).

### Dwarf Planets



Pluto



Eris



Ceres





## IN ADDITION TO ASTEROIDS... THERE ARE COMETS AND METEOROIDS TOO!

### COMETS

- These are balls of rock and ice that grow tails as they approach the Sun, in the course of their highly elliptical orbits.
- As comets heat up, gas and dust are expelled and trail behind them.
- The Sun illuminates this trail, causing it to glow.



### METEOROIDS

- Meteoroid is a small particle from an Asteroid or Comet.
- Lumps of rock or iron that orbit the Sun.
- Sometimes these, don't vaporize completely in the atmosphere.



## LONAR LAKE



This is the lake formed when a **Meteorite** hit India at a place Maharashtra around 52,000 years ago.

**DIFFERENCES FOR CRYSTAL CLEAR CLARITY****ASTEROIDS**

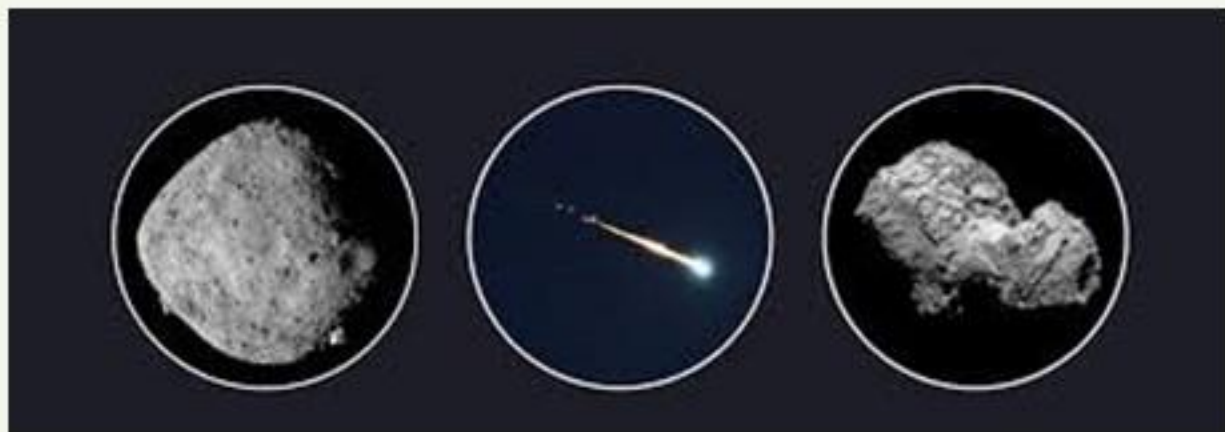
- Numerous rocky and airless leftovers.
- Move around the Sun in elliptical path.

**COMETS**

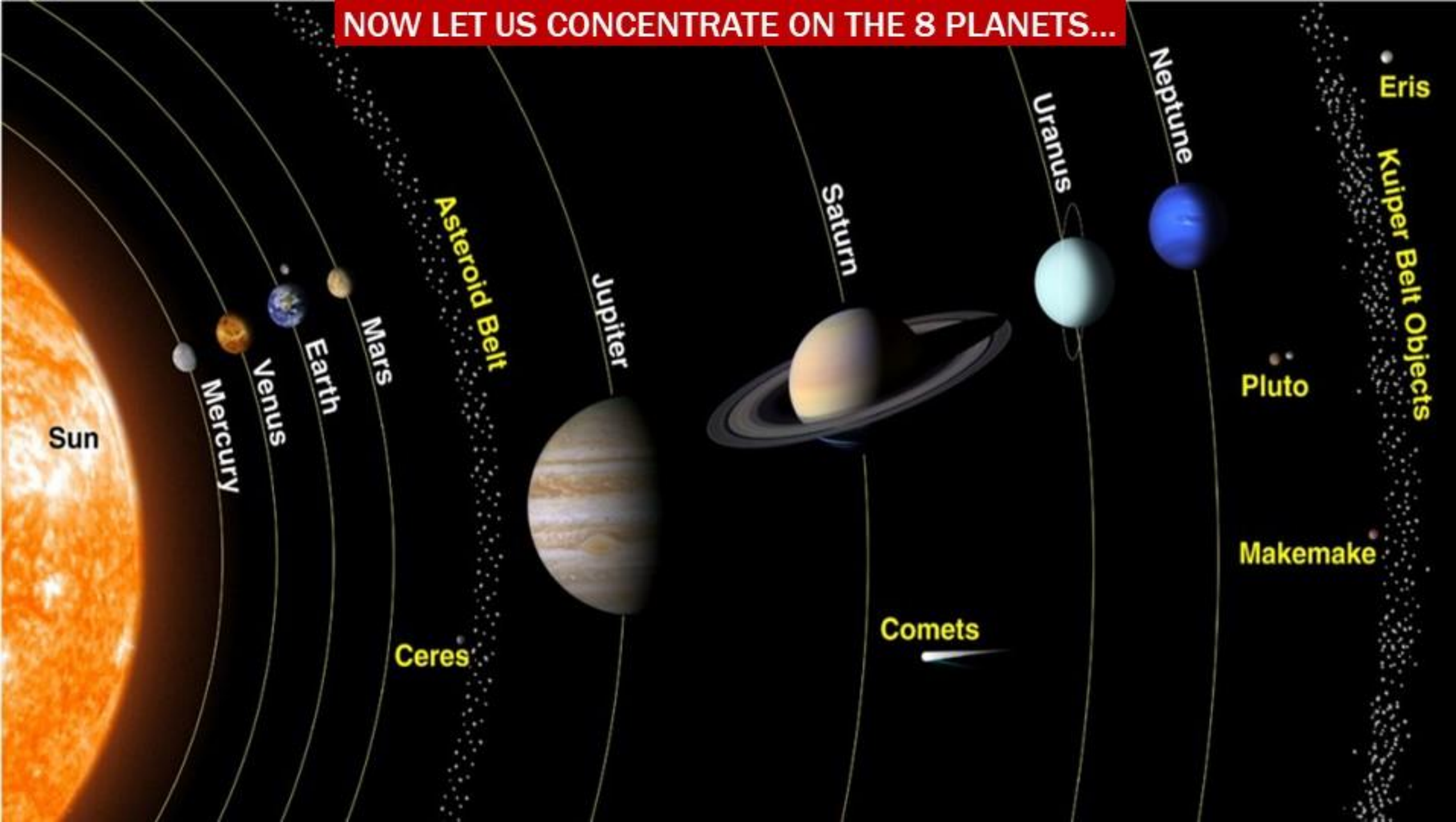
- Dirty space snowballs of mostly ice and dust.
- Highly elliptical orbits than asteroids.

**METEOROIDS**

- The true space rocks of the solar system.
- Move around the Sun.



NOW LET US CONCENTRATE ON THE 8 PLANETS...



**CLASSIFICATION OF THESE 8 PLANETS****1. TERRESTRIAL PLANETS**

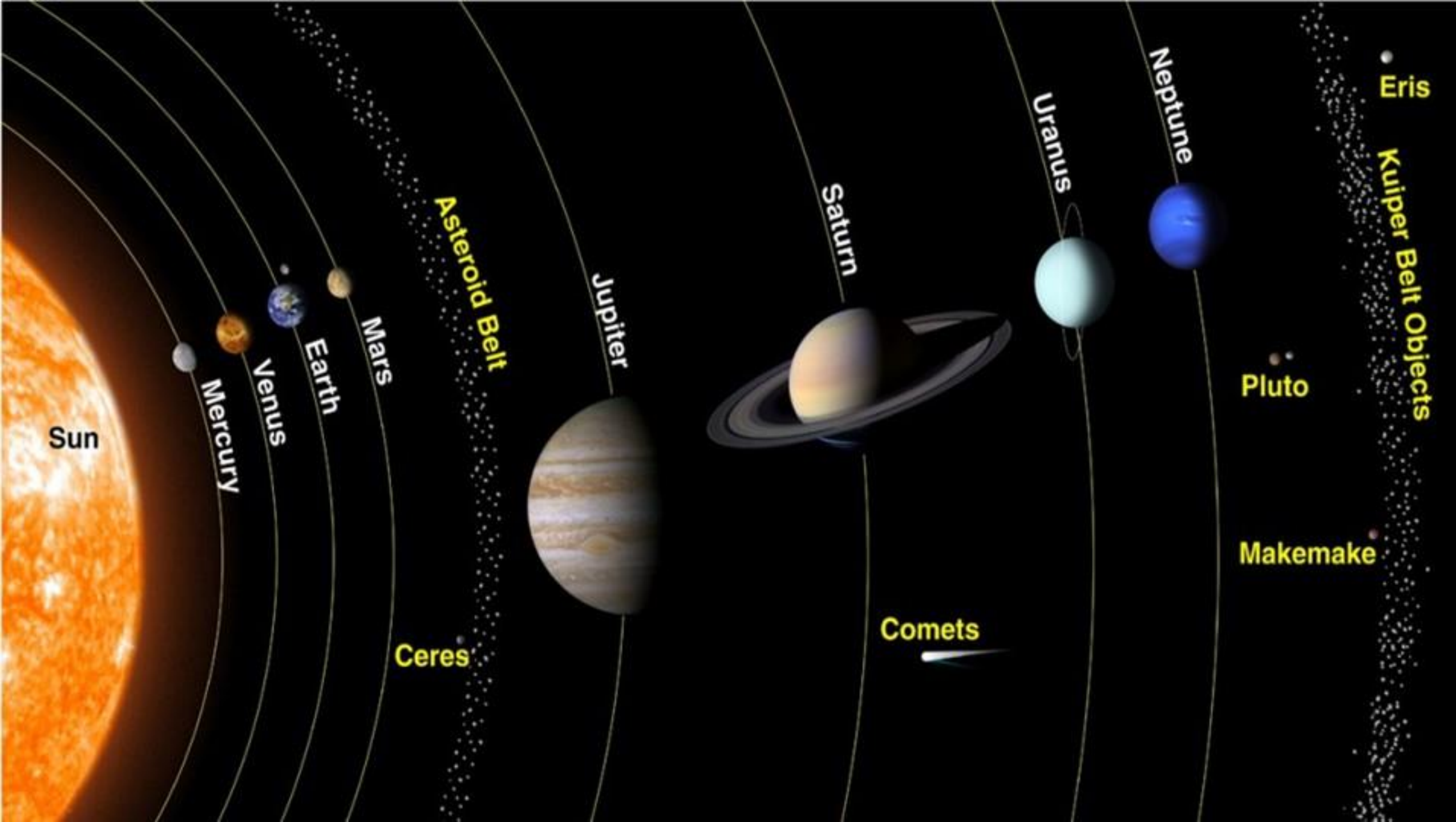
- These were formed in the close vicinity of the parent star.
- The solar wind is more intense.
- Smaller and their lower gravity could not hold the escaping gases.

**2. JOVIAN PLANETS**

- These were formed at quite a distant location.
- The solar winds are not that intense.
- Larger in size when compared to the terrestrial planets.



**Solar wind:** It is the stream of charged particles released from the upper atmosphere of the Sun.



| Mercury<br>8 <sup>th</sup>                                | Venus<br>6 <sup>th</sup>      | Earth<br>5 <sup>th</sup> | Mars<br>7 <sup>th</sup>            | Jupiter<br>1 <sup>st</sup>                | Saturn<br>2 <sup>nd</sup> | Uranus<br>3 <sup>rd</sup>              | Neptune<br>4 <sup>th</sup> |
|-----------------------------------------------------------|-------------------------------|--------------------------|------------------------------------|-------------------------------------------|---------------------------|----------------------------------------|----------------------------|
| INNER PLANETS                                             |                               |                          |                                    | OUTER PLANETS / JOVIAN PLANETS (99% mass) |                           |                                        |                            |
| Closely spaced Rocky and Terrestrial Silicates and Metals |                               |                          |                                    | Gas Giants<br>Hydrogen & Helium           |                           | Ice Giants<br>Water, Ammonia & Methane |                            |
| No Moon                                                   | No Moon                       | 1 Moon                   | 2 Moons                            | 69 Moons                                  | 61 Moons                  | 27 Moons                               | 14 Moons                   |
| Smallest                                                  | Hottest<br>@462°C             |                          |                                    | Largest & Heaviest                        |                           |                                        |                            |
|                                                           | Brightest planet              | Blue Planet              | Red Planet                         | Has Rings                                 | Has Rings                 | Has Rings                              | Has Rings                  |
|                                                           | Morning Star/<br>Evening Star | Supports Life            | Highest Mountain<br>(Olympus Mons) | Lord of the Heavens                       |                           |                                        |                            |
|                                                           | Earth's Twin                  |                          |                                    |                                           |                           |                                        |                            |
| 88 Days                                                   | 225 Days                      | 365 Days                 | 688 Days                           | 11.9 Years                                | 29.5 Years                | 84 Years                               | 164.8 Years                |

Inner and outer planets are separated by Asteroid belt



## VENUS HAS MANY NICKNAMES!

### MORNING STAR/EVENING STAR

As the Earth rotates from west to east, for an observer on the Earth, the Venus would rise before the Sun. For the Sun to rise, the Earth would have to rotate further by some degrees.



### BRIGHTEST PLANET

- It is due to high *Albedo*.
- Albedo is nothing but ratio of the *reflected radiation* to the *incident radiation*.
- It has albedo value of **0.75**.

### EARTH'S TWIN

Venus is referred to as Earth's twin because both the planets are almost alike in size, mass, density and gravity.





## SEVERAL MISSIONS TO PROBE VENUS AND MARS!

### VENUS MISSION

During the 2028-2030, NASA plans to launch an Orbiter called '**VERITAS**', and an atmospheric probe known as '**DAVINCI+**' with price tag of each mission capped at around \$ 500 Million.

### THREE MARS MISSIONS

- AL-Amal (HOPE) by UAE.
- Tianwen-1 (Quest for Heavenly Truth) by China.
- Perseverance Rover by NASA.





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WE EXPRESS OUR  
SINCERE THANKS  
FOR VIEWING THIS VIDEO

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